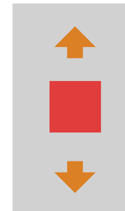
Movement 07

In the last lesson, we learnt about game design. To play a game, we need the objects in it to move. In this lesson, we will look at how to apply movements to a game. Up or down movements are called vertical movements and left, or right ones are called horizontal movements. In a two-dimensional (2D) space, only these types of movements are possible.

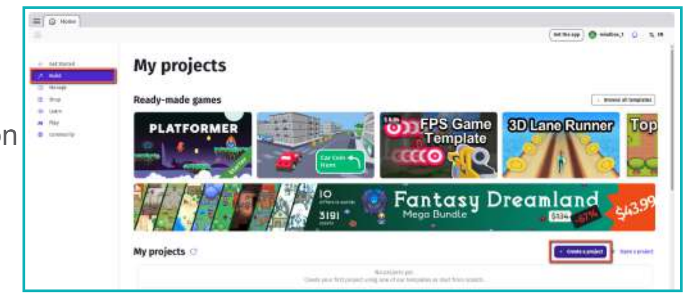
7. Movement

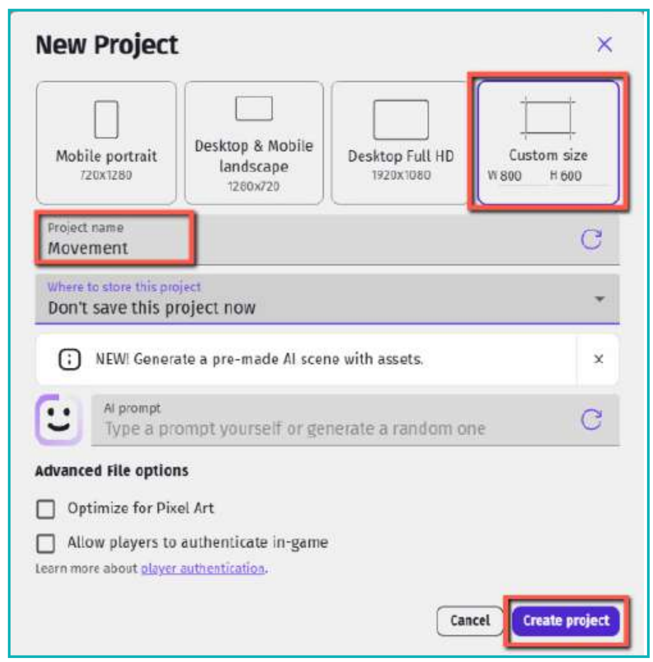
We will learn how to add movement to an object, like in the Pong Game. The object should move when a key is pressed on the keyboard.



To add movements, we will use Gdevelop. The interface of Gdevelop looks like this:

1 To start creating our first game, choose to click on the ‘create a project’ option.

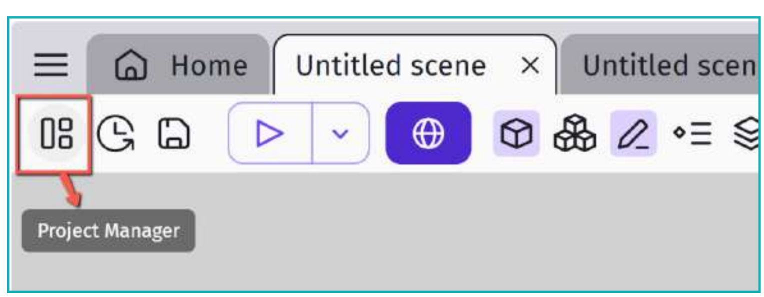




2 We have a choice of the game resolutions listed above. Choose a name for the project and save it to your computer. Ensure that the “authenticate in-game” option is not selected.

A new window will open. In a game, we need to create multiple levels, inventory, and screens. The Scene option lets us create all of these. So, our first step is to create a Scene.

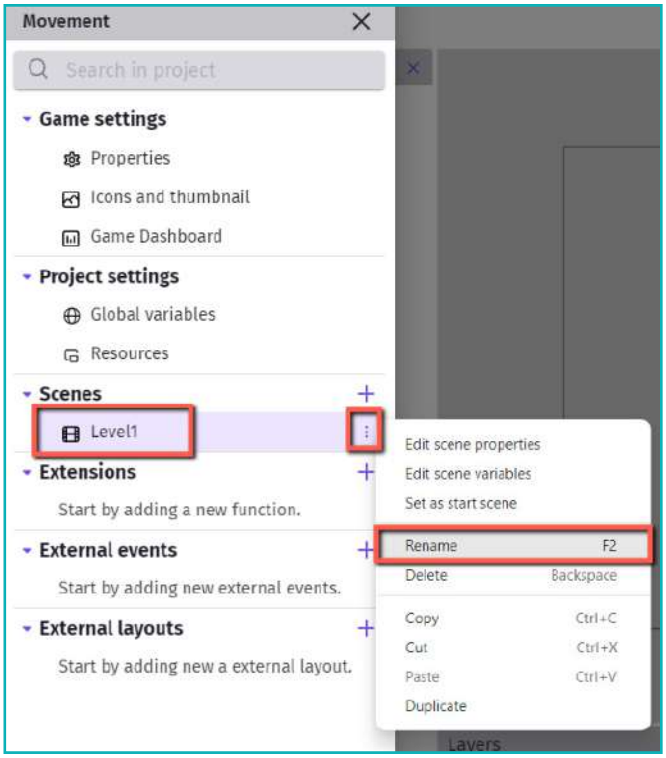
3 Navigate to the upper left corner and select the project manager option. Scenes and other game settings menus can be accessed from here.



We need to give the scene a name that suits the game.

The default untitled scene and event sheet are already available. We now need to rename the scene.

Note: It is a good practice to name scenes and assets. This helps us stay organized when we create the game.



4 Click on the three dots and select "Rename". Type a name for the scene.

Exercise

Answer in one line

1. What is a Scene?

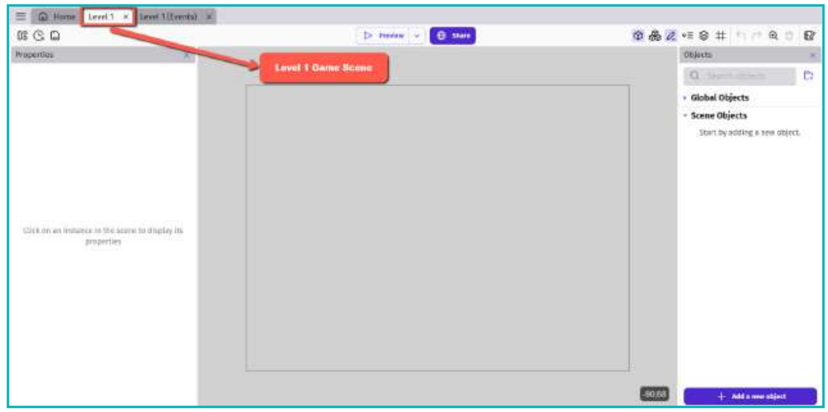
____

____

2. How do you rename a scene?

____

____



5 Double click on the scene and the Level1 scene will open.

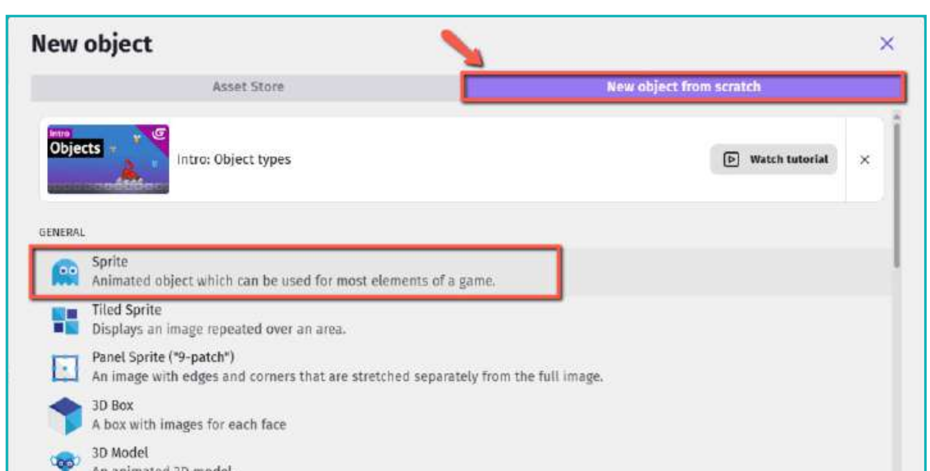
We need to add different elements in the game. Any visible or invisible element in the game such as players, obstacles, or enemies are called objects.

6 To create an object, click on Add a new object in the Objects panel.

Objects are of different types. For now, we will select Sprite.

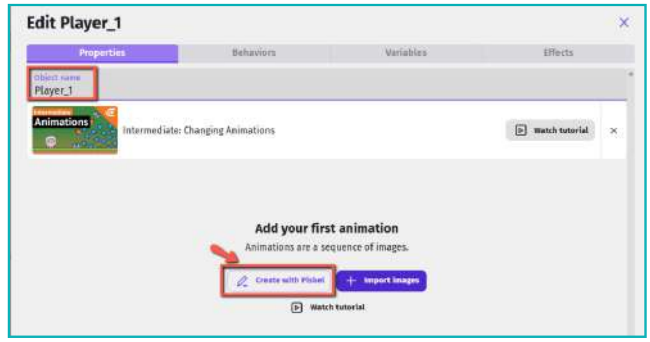
Sprites

A sprite is a two-dimensional image. Sprites are a visual representation of an object.

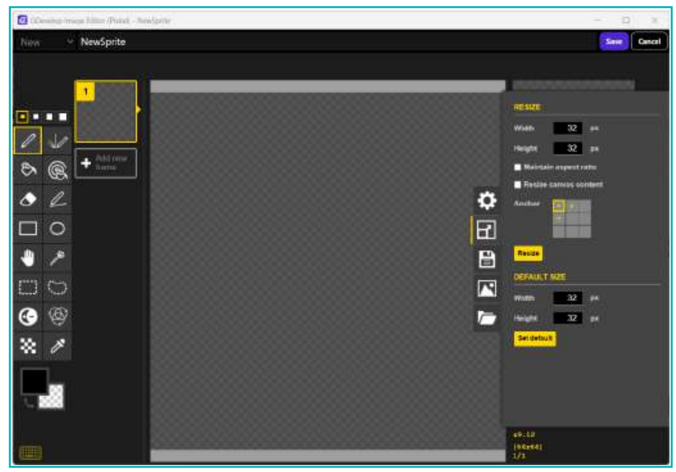


1. Name the object.

2. To define the look of a sprite, click on create with piskel.



3. This will open the editor window, where we can design the sprite.

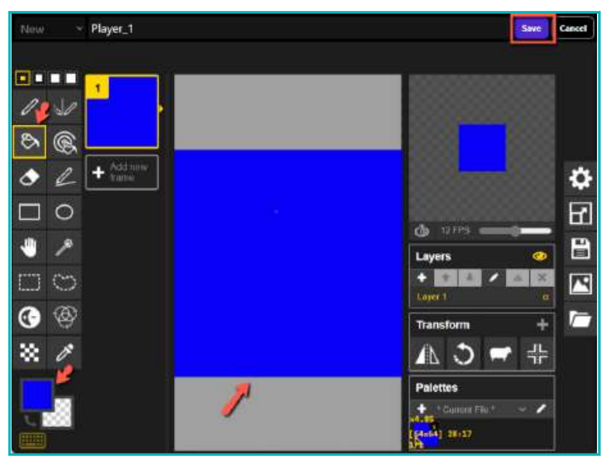


4. Fill colour into the object.

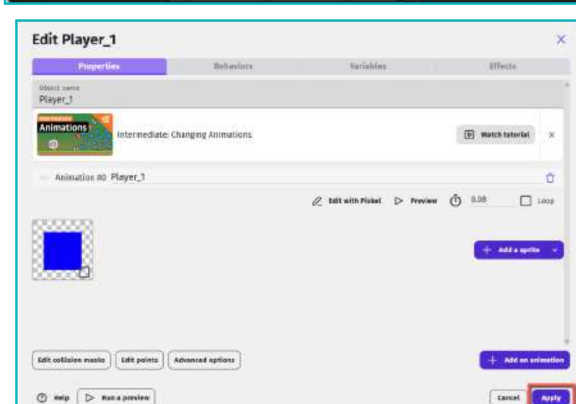
Note: We will create a sample object with a square shape and red colour.

5. Use the paint bucket tool.

6. Click on save to save the designed sprite.

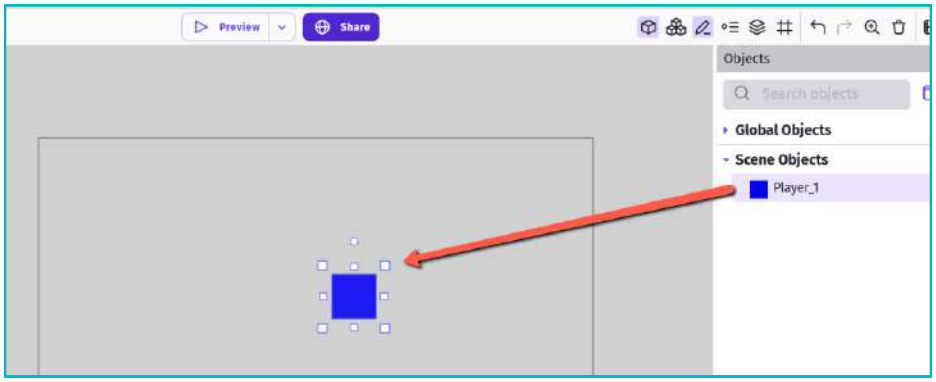


7. Click on the APPLY button.



8 The sprite we just designed will appear in the objects panel.

9 Click and drag it to the scene area.



Apply Movement

In game design, instructions are given through events and actions.

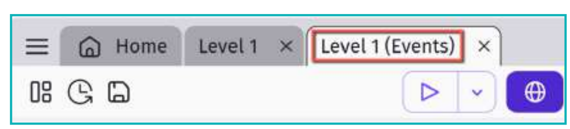
Note:

Events: Events are logical conditions that would occur during the runtime.
For example: pressing a button.

Actions: Actions take place after an event.

All events and actions for a game made in GDevelop are placed in the event sheet panel.

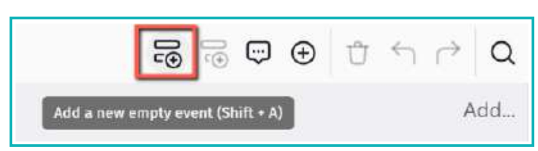
1 Click on the event sheet "Level1(Events)"



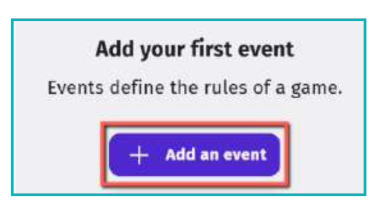
We need to decide movements of an object. If the up-arrow key is clicked, then the object should move up. If the down arrow key is pressed, then the object should move down. For this, we will create a new event.

There are multiple ways to create a new event:

Method 1: Click on "Add a new empty event".

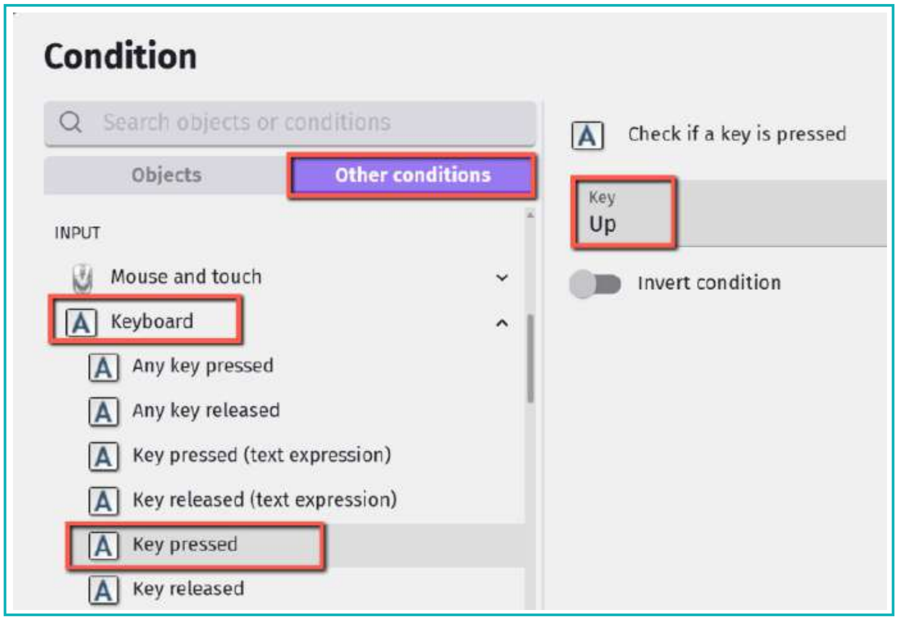


Method 2: Click on "Add an event".



2 Once an empty event is created, we will define the conditions that need to be met for movements.

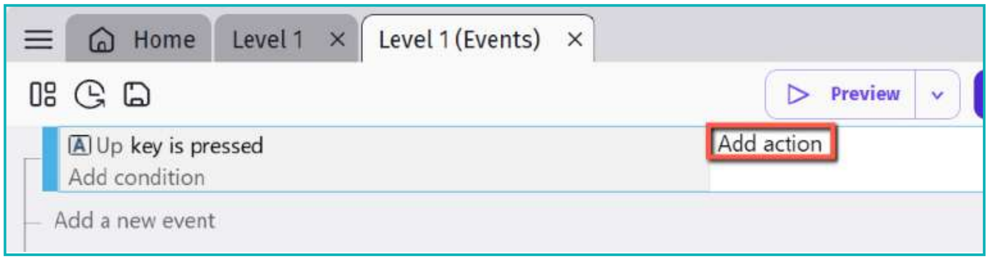
3 In our case, we want the object to move when arrow keys are pressed.



4 Now, we need to set an action.

5 The action will decide what we expect when the up key is pressed.

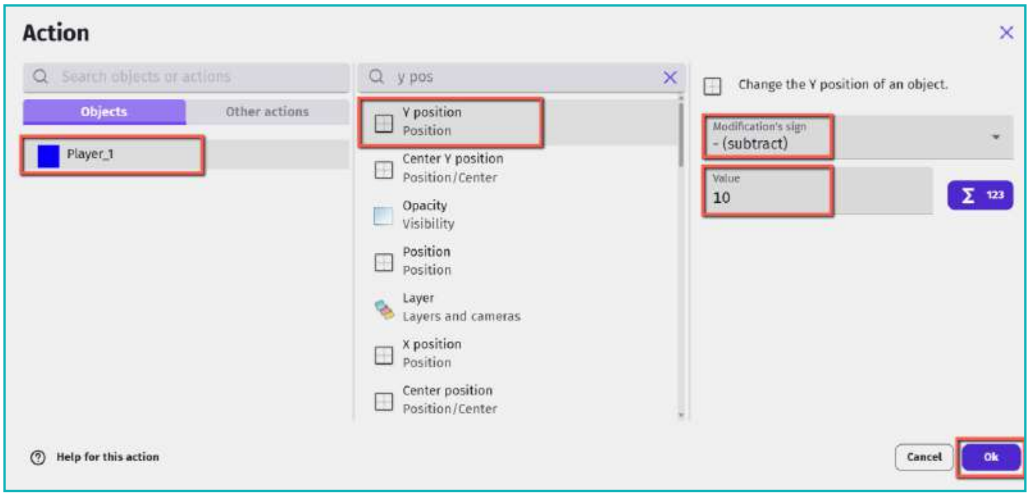
6 Click on Add action.



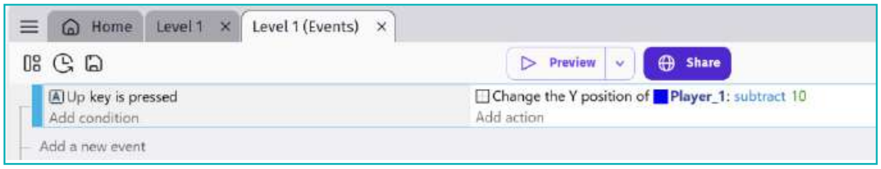
In a game, every object placed on the scene is assigned x and y coordinates. These coordinates determine the object's horizontal position (X-axis) and vertical position (Y-axis).

The X-axis values decrease as you move to the left and increase as you move to the right. On the other hand, the Y-axis values increase as you move downwards and decrease as you move upwards.

Note: Try out different values, depending on the speed at which you want the object to move.

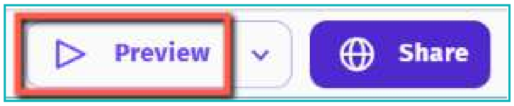


The event should look like this:



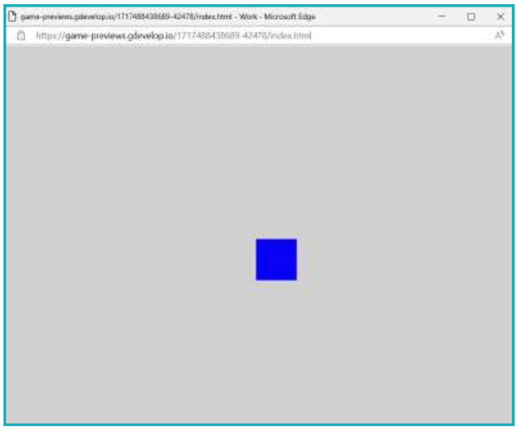
Let's test this out to see if it works.

7. Click on the Preview button in the top menu.

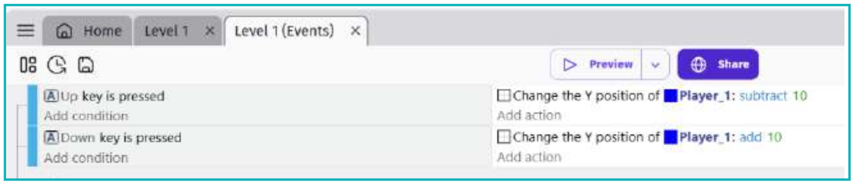


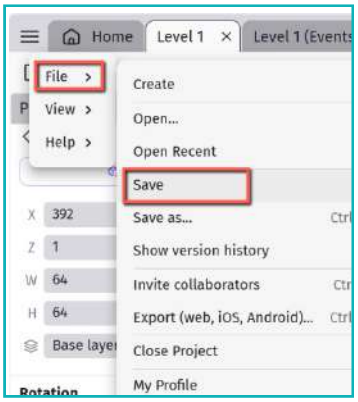
8. A project preview window will pop up.

9. When we press the up-arrow key, the object should move upward.



Similarly, add another event to move the object downward.

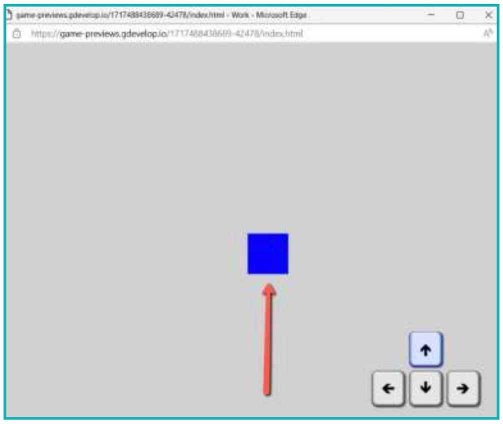




10 Test the movement and save the game.

11 We can name it “Movement”.

12 The final output should look something like this:



Every 2D or 3D game has at least one moving object. The player controls the object’s movements to try and win the game. We’ve learnt how to create an object and give it movements. Now, we can create other objects which can move and be controlled with a keyboard.

Exercise

Fill in the blanks

3 ________________ are visual representations of an object.

4 All events and actions for a game created in GDevelop are placed in the ________________ panel.

5 The ________________ button in the Top Menu is used to test the game.

Tech Flash

- **Sprites** : Sprites are 2 dimensional images.
- **Events** : Events are logical conditions that occur during the runtime.
- **Actions** : Actions are commands that will be executed on occurrence of an event.
- **Objects** : Any visible/invisible element in the game like players, obstacles, and enemies are termed objects.
- **Scene** : It is the area where game levels are designed and built.